**Title:** Data Profiler
**Duration:** 5 Hours

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## 🎯 Objective

You are tasked with conducting **Data Preprocessing and Feature Engineering** on a real-world dataset. The aim is to **understand, clean, transform, and analyze** the dataset before it can be used for machine learning. This project emphasizes **data profiling, handling multiple formats, and performing EDA**.

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## 📑 Problem Statement

You have been hired as a **Junior Data Analyst** by a consumer insights company. The company has provided a dataset containing **customer purchase behavior** collected from multiple sources (CSV, JSON, SQL database, and an API). The goal is to **frame a machine learning problem** (predict customer churn) and perform **data preprocessing and profiling** to make the dataset ML-ready.

You must apply the concepts of **data analysis, tensors, data cleaning, and exploratory data analysis (EDA)** to extract insights.

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## 📝 Tasks

### Part A: Fundamentals

1. Write a short note: *What is Data Analysis?*
2. Describe how to plan a **Data Science Project**, listing all steps.
3. Frame a **Machine Learning problem statement**: Predict whether a customer will churn based on purchase behavior.

4. Explain:
    - What are **Tensors**?
    - Provide an **in-depth explanation** of tensors with simple examples using NumPy.

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**Part B: Data Acquisition**

5. Import datasets from different sources:
    - Load CSV file using Pandas.
    - Parse a JSON file.
    - Connect to a SQL table and fetch records.
    - Fetch data from an API (e.g., random user API or dummy REST API).

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**Part C: Data Understanding & Cleaning**

6. Perform **initial exploration**:
    - Use `.head()`, `.info()`, `.describe()` to explore.
    - Identify missing values and duplicates.

7. Apply **data cleaning**:
    - Handle missing data (imputation or removal).
    - Correct inconsistent data types.
    - Drop irrelevant columns.

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**Part D: Exploratory Data Analysis (EDA)**

8. Perform **Univariate Analysis**:
    - Distribution plots of Age, Income, and Purchases.

9. Perform **Bivariate Analysis**:
    - Relationship between Gender & Purchases.
    - Relationship between Income & Churn.

10. Perform **Multivariate Analysis**:
    - Correlation heatmap of all numerical variables.
    - Pair plots to identify feature interactions.

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**Part E: Data Profiling**

11. Generate a **Pandas Profiling Report** that summarizes:
    - Missing values.
    - Descriptive statistics.
    - Correlations.
    - Warnings on potential data quality issues.

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## 📌 Expected Outcome

By completing this project, students will:

- Gain hands-on practice in **loading and merging datasets** from multiple sources.
- Understand the **role of tensors** in machine learning.
- Perform **data cleaning and preprocessing**.
- Carry out **EDA (Univariate, Bivariate, Multivariate)** with visualizations.
- Generate **automated profiling reports** for decision-making.
- Frame a **machine learning problem** from raw data.

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### ✅ Submission Guidelines

- Include practical implementation in a Jupyter Notebook and screenshots.
- Label all charts clearly and write short interpretations under each result.
- **GitHub Repository:**
    - Create a GitHub repository to host your project.
    - Upload your project files, including source code, and documentation to the repository.
    - Add a document (PDF) explaining theory concepts with definitions (if any).
    - Ensure that you provide a clear and descriptive README.md file.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!

**Data Profiler**

Data Preprocessing and Feature Engineering

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**BRING ON YOUR CODING ATTITUDE**